Integer solutions to linear programming applications (Extended)

ANSWER KEY

Core Concepts: Integer solutions to linear programming applications (Extended)

1 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(2)$.

a. 13 ✓

b. 18

c. 10

d. 17

Substitute $x = 2$: $f(2) = 5(2) + 3 = 13$.

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

3 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

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Procedural Fluency & Algebraic Manipulation

4 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(3)$.

a. 18 ✓

b. 23

c. 15

d. 22

Substitute $x = 3$: $f(3) = 5(3) + 3 = 18$.

5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

6 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

Graphical & Geometric Interpretation

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7 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(4)$.

a. 23 ✓

b. 28

c. 20

d. 27

Substitute $x = 4$: $f(4) = 5(4) + 3 = 23$.

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

9 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(5)$.

a. 28 ✓

b. 33

c. 25

d. 32

Substitute $x = 5$: $f(5) = 5(5) + 3 = 28$.

11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ **and** $x = 9$ ✓

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

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